CAPSTONE TECHNICAL ENGINEERING REPORT

ATHENAEUM AI 2.0: AN AUTONOMOUS MULTIMODAL ACADEMIC RESEARCH AND SPATIAL-SEMANTIC LIBRARY INTELLIGENCE SYSTEM

A Modular, Production-Oriented Architecture Integrating Dual-Channel Reciprocal Rank Fusion ($k=60$), Omnichannel Multimodal LLM Disambiguation, WebGL Neural Explorers, and ACID Concurrency Guarantees

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DECLARATION

I, **Aryan Dutta**, hereby declare that the technical capstone project titled **“ATHENAEUM AI 2.0: An Autonomous Multimodal Academic Research and Spatial-Semantic Library Intelligence System”** submitted to the Department of Computer Science and Engineering for the course **CSE 276**, is an original record of engineering work executed under rigorous academic and software engineering standards.

All algorithmic implementations—including the Dual-Channel Reciprocal Rank Fusion ($k=60$) search engine, the 384-dimensional dense cosine vector similarity layer, the LangGraph multi-agent copilot StateGraph pipeline, the multimodal document analysis pipeline, and the SQLite transactional concurrency isolation mechanisms—were designed, implemented, benchmarked, and verified in accordance with university ethical codes and production-oriented engineering methodologies. Where external libraries, datasets (OpenAlex, arXiv, Crossref), or architectural frameworks were integrated, they have been formally cited and attributed within the methodology and references section.

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_____

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Detailed structural roadmap and page index for CSE 276 technical evaluation

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i) Introduction (including objective)

1.1 Problem Context & Contemporary Academic Information Retrieval Gaps

Academic research environments in contemporary higher education institutions represent highly complex, multidisciplinary ecosystems. In a traditional university library setting, scholars, graduate researchers, and undergraduate students are confronted with severe structural friction across three fundamental operational domains: (1) *Fragmented Resource Discovery*, where physical monograph holdings, electronic reserve materials, and global open-access scholarly repositories reside in disparate, mutually unintelligible silos; (2) *Lexical Brittleness in Search Engines*, which demand exact keyword and Boolean syntax matches, thereby failing catastrophically when patrons articulate cross-disciplinary inquiries, whiteboard diagrams, or conceptual research paradigms; and (3) *Spatial Opacity*, wherein the physical study spaces, departmental study pods, and dynamic library occupancy remain entirely decoupled from the digital circulation ledger.

Legacy Integrated Library Systems (ILS), such as Koha, Sierra, or Voyager, were architected decades ago during an era dominated by relational indexing over rigid bibliographic standard formats (e.g., MARC21, ISO 2709). While effective for deterministic bibliographic record-keeping, these traditional engines exhibit fundamental architectural limitations when dealing with unstructured research inputs: handwritten equations, vector embeddings, architectural schematics, audio lecture recordings, or multimodal research documents. Furthermore, the modern academic landscape has witnessed an explosive growth in digital scholarship, with platforms like OpenAlex (indexing 250M+ scholarly works across journal articles, books, theses, and preprints), Cornell arXiv (2.4M+ preprints), and Crossref (150M+ metadata records for registered DOIs). Because records across these external platforms partially overlap, the library retrieval engine must federate across these repositories while applying cross-provider DOI and title deduplication.

Modern computational pedagogy demands a unified platform capable of understanding the user's conceptual intent. When a student reads a paper on *'hierarchical nanorobotic control architectures'*, they should not be required to manually decompose the inquiry into disjointed keyword queries such as *'nanorobots'*, *'biomedical systems'*, or *'distributed control'*. The library platform must autonomously parse the complex semantic interdependencies, identify relevant monographs in the physical stacks, recommend foundational literature, and retrieve contemporary peer-reviewed articles and preprints with zero manual overhead.

1.2 System Engineering Objectives & Core Requirements

To decisively overcome the architectural inertia of conventional library platforms, **Athenaeum AI 2.0** was formulated, engineered, and deployed as an enterprise-grade academic research intelligence and space-management operating system. The project is governed by five mandatory engineering objectives:

1. **Dual-Channel Hybrid Retrieval with Reciprocal Rank Fusion (RRF $k=60$):** Synthesize high-dimensional dense vector embeddings (384-dimensional latent semantic space generated via sentence-transformers/all-MiniLM-L6-v2) with high-precision lexical token matching (BM25 with domain-specific tokenization). The system must deterministically merge lexical and semantic rank lists to eliminate vocabulary mismatch while preventing zero-match keyword queries.
2. **Omnichannel Multimodal Research Disambiguation:** Ingest heterogeneous academic media—including whiteboard diagrams, LaTeX mathematical equations, audio recordings, PDF research papers, PowerPoint slide decks (.pptx), and Word documents (.docx)—via an intelligent multi-engine selector (supporting Groq Llama 3.3 70B, OpenAI GPT-4o, Anthropic Claude 3.5 Haiku, Google Gemini 1.5, and zero-cost local Python fallbacks). The system must extract core concepts, sanitize search keywords, and federate queries across campus monographs and 250M+ scholarly works indexed by OpenAlex, arXiv, and Crossref with automatic DOI deduplication.
3. **Autonomous Multi-Agent StateGraph Copilot:** Implement an autonomous library agent capable of multi-step tool execution (catalog discovery, book renewal, reserve validation, 3D WebGL neural scene invocation) via a cyclical state graph with full action tracing, latency verification, and academic disambiguation.
4. **Spatial-Semantic Wayfinding & IoT Telemetry:** Model physical library floors using planar Cartesian coordinates (x, y, z). Implement Euclidean clustering, multi-floor wayfinding graphs, and discrete Kalman filtering over live telemetry streams to provide real-time occupancy heatmaps, quiet-zone detection, and study pod reservation locks.
5. **ACID Concurrency & High-Assurance Operational Resilience:** Enforce transactional integrity across high-concurrency circulation desks using immediate writer locks via BEGIN IMMEDIATE in SQLite WAL mode, serializing competing write transactions during peak monograph borrowing. The platform passed all 423 automated tests in the recorded test run.

1.3 Architectural Vision: From Static Catalogues to Autonomous Agentic Hubs

Athenaeum AI 2.0 fundamentally transitions the university library from a passive book warehouse to an active, agentic research hub. Rather than forcing students to decipher Dewey Decimal numbers or guess exact catalog titles, the platform acts as an intelligent academic co-pilot. A student analyzing a complex computational concept (such as the Transformer self-attention mechanism) can verbally query the copilot, upload an architectural diagram, interact with a live 3D WebGL neural simulation, discover physical monographs on Floor 3 with exact shelf coordinates, and retrieve matching peer-reviewed literature and preprints across OpenAlex and arXiv.

This architectural vision is underpinned by three core design maxims: *Low-Latency User Experience*, where client-side state is preserved and cached to reduce visible layout shifts; *Decoupled Provider Resiliency*, ensuring that cloud provider deprecations or outages do not disrupt core library workflows; and *Rigorous Epistemological Grounding*, whereby AI agents are strictly prohibited from inventing fake academic citations or non-existent book inventory.

1.4 Comparative Paradigm Matrix: Traditional OPAC vs. Athenaeum AI 2.0

The following analytical matrix formally contrasts the structural and computational capabilities of conventional library software (such as Koha, Evergreen, and standard Web-OPACs) against the autonomous capabilities engineered in Athenaeum AI 2.0:

Architectural Dimension	Traditional Integrated Library Systems (ILS)	Athenaeum AI 2.0 Autonomous Platform
Retrieval Paradigm	Strict keyword/Boolean matching (SQL LIKE, inverted index); brittle to typos and synonyms.	Dual-Channel Reciprocal Rank Fusion ($k=60$); merges BM25 lexical tokens with 384-dim dense vectors.
Multimodal Ingestion	Completely absent; manual cataloguing of bibliographic MARC21 records.	Omnichannel ingestion: PDFs, Word, Slides, audio clips, handwritten LaTeX equations, and diagrams.
External Discovery	Siloed to internal physical holdings; external databases require separate federated search.	Unified federation across local catalog, OpenAlex (250M+ works), arXiv preprints, and Crossref DOIs with deduplication.
Spatial Intelligence	Static textual shelf location strings (e.g., 'Stack 4B, Shelf 2'); no indoor routing.	Cartesian coordinate mapping (x,y,z), Euclidean clustering, A* pathfinding, and live 3D WebGL exploration.
Occupancy Telemetry	Periodic manual headcounts; static occupancy assumptions.	Real-time IoT telemetry ingestion; discrete Kalman filtering; dynamic quiet-zone detection.
Circulation Concurrency	Relies on database connection pools; vulnerable to contention under peak surges.	ACID transactions with SQLite WAL mode and BEGIN IMMEDIATE writer lock serialization.

ii) Project Gap / Scope

2.1 Critical Vulnerabilities & Limitations of Legacy Integrated Library Systems

A rigorous empirical analysis of contemporary academic libraries reveals severe technological vulnerabilities that impair research productivity. Traditional ILS architectures suffer from three fatal design flaws:

1. **The Semantic Gap & Vocabulary Mismatch Problem:** Standard lexical search engines rely on exact token overlap. If a student queries *'generative adversarial nets'*, an exact match engine fails to rank authoritative books entitled *'Deep Generative Models'* or *'Synthetic Neural Sampling'* unless explicit manual cross-references exist. This creates artificial research barriers.
2. **Monolithic Relational Fragility:** Traditional OPACs run complex multi-table SQL joins across normalized catalog tables on every keystroke. Under peak campus traffic (such as final exam registration), database connection pools saturate, resulting in high query latencies and service lockups.
3. **Zero Spatial-Physical Integration:** Even when a student locates a book record in an OPAC, the system provides no spatial guidance. The patron is left to wander multi-story library stacks searching for Dewey Decimal numbers, with zero visibility into whether adjacent study pods or quiet desks are currently occupied.

2.2 Functional Scope Boundaries & System Operational Envelope

To ensure production-oriented engineering feasibility and high reliability within university computing infrastructure, the functional scope of Athenaeum AI 2.0 is rigorously bounded by the following operational envelope:

- **Internal Catalog Management:** High-speed indexing and vector embedding of 1,500 core academic monographs across 10 disciplines (Computer Science, Electrical Engineering, Physics, Mathematics, Biology, Chemistry, Mechanical Engineering, Economics, Philosophy, and Literature).
- **Live External Scholarly Repositories:** Real-time federated query execution against OpenAlex, Cornell arXiv, and Crossref REST APIs, retrieving verified scholarly articles, preprints, author metadata, publication dates, and open-access download URLs.
- **Spatial Library Modeling:** Precision Cartesian mapping across four architectural tiers (Ground, 1st, 2nd, and 3rd floors) comprising 12 study rooms, 60 study pods, 4 service zones, and 24 primary book stack ranges.
- **Real-Time Telemetry & Environmental Sensing:** Simulation and ingestion of acoustic decibel streams, infrared motion sensors, and seat occupancy gates filtered through discrete Kalman estimation algorithms.
- **Concurrency & Multi-Tenancy:** Sub-50ms transaction handling supporting concurrent patron sessions under SQLite Write-Ahead Logging (WAL) mode.

2.3 Empirical Failure Modes & Academic User Journey Case Studies

To ground the system design in genuine academic workflows, three canonical failure modes of existing systems were analyzed:

Case Study A — The Cross-Disciplinary Student Inquiry: An undergraduate researcher in computational neuroscience requires literature on how recurrent neural networks simulate hippocampal place cells. Searching an existing library OPAC for *'hippocampal place cells in recurrent networks'* yields 0 results because no single book title contains that exact phrase. In Athenaeum AI 2.0, dense vector similarity maps the inquiry directly to relevant monographs in both Neural Computation and Cognitive Biology, while the external aggregator fetches the latest peer-reviewed works and arXiv preprints via OpenAlex.

Case Study B — The Multimodal Whiteboard Encounter: A graduate student photographs a professor's whiteboard diagram depicting a distributed consensus state machine with LaTeX equations ($f < \frac{n}{3}$). Traditional library systems offer zero ingestion capability for visual media. Athenaeum AI's Multimodal Omnichannel Ingestion parses the diagram, extracts the Byzantine Fault Tolerance formulation, queries the internal catalog for distributed systems monographs, and routes the student directly to Shelf Range 2B on Floor 2.

Case Study C — The Exam-Week Concurrency Crunch: During midterm week, dozens of students simultaneously attempt to reserve study pod SP-102. In legacy systems lacking serialized transactions, race conditions result in multiple double-bookings and student friction. Athenaeum AI's SQLite BEGIN IMMEDIATE atomic lock serializes checkout requests, granting the reservation to the first connection and immediately returning an informative HTTP 409 Conflict with suggested alternative pods.

2.4 Comprehensive Edge Case Analysis & Fail-Safe Mechanics

Enterprise software requires defensive programming and deterministic failure recovery. Athenaeum AI incorporates comprehensive fail-safe mechanics for critical edge cases:

1. **High-Concurrency Monograph Contention:** If multiple patrons simultaneously attempt to borrow the last physical copy of a monograph (e.g., during exam week), standard read-modify-write patterns result in race conditions. Athenaeum AI implements an atomic_write() context manager that executes BEGIN IMMEDIATE under SQLite WAL mode, acquiring the database writer lock before reading inventory and updating copy counts. Competing write transactions are serialized, and all contention tests pass with zero negative inventory.
2. **Upstream AI Provider Outages & Rate Limiting:** When patron inquiries exceed external LLM rate limits (HTTP 429) or upstream cloud models experience outages, the multimodal coordinator automatically falls back through a deterministic cascade: Groq Fast (Llama 3.3 70B) → OpenAI GPT-4o → Anthropic Claude → Google Gemini → Local Python Document Engine. The local engine extracts structured text from documents using pypdf, python-docx, and python-pptx with zero external network dependencies, ensuring basic discovery remains functional offline.
3. **Corrupted or Zero-Length Vector Embeddings:** If an imported monograph possesses an unparseable vector string or malformed dimension, the hybrid search engine gracefully isolates the exception, logging a warning while continuing search execution using pure lexical BM25 matching without system crashes.
4. **Telemetry Sensor Glitches & Outliers:** Physical IoT sound meters occasionally register spurious instantaneous acoustic spikes (e.g., dropping a heavy book). The discrete Kalman filter dynamically calculates the optimal Kalman gain K_k, smoothing spurious transients and preventing false 'Loud Noise' alarms in designated Quiet Study Zones.

iii) Methodology (including library & uses, mathematical equations)

3.1 Layered Distributed Systems Architecture

Athenaeum AI 2.0 is structured as a decoupled, multi-tier asynchronous architecture comprising four primary layers: (1) *Patron & Staff Client Layer* (React 19 SPA, TypeScript, Three.js WebGL canvas, Vite bundling); (2) *Application & API Gateway* (FastAPI asynchronous microservice framework, Pydantic v2 validation models, JWT security); (3) *Intelligence & Semantic Services* (LangGraph state pipelines, sentence-transformers neural encoders, multimodal vision routers); and (4) *Persistence & Storage Layer* (SQLAlchemy ORM, SQLite WAL mode, in-memory caching layers).

Communication between layers occurs strictly via strongly typed data contracts defined in Pydantic v2 schemas. The asynchronous ASGI event loop guarantees non-blocking I/O across outbound academic API queries, internal vector calculations, and spatial coordinate transformations.

3.2 Dual-Channel Reciprocal Rank Fusion (RRF $k=60$) Mathematical Model

To achieve optimal search performance across both structured bibliographic metadata and unstructured natural language queries, Athenaeum AI implements Dual-Channel Reciprocal Rank Fusion (RRF). RRF solves the classic problem of score normalization: cosine vector similarity produces bounded real numbers in the range $[-1.0, 1.0]$, whereas lexical BM25 produces unbounded positive scores. Directly summing these heterogeneous scores requires fragile hand-tuned scalar weights that degrade across varying document lengths.

Instead of merging raw scores, Reciprocal Rank Fusion ranks candidate documents independently in each channel, then computes a fused score based solely on their reciprocal rank positions:

$$\text{Score}_{\text{RRF}}(d) = \sum_{m \in M} \frac{1}{k + r_m(d)}$$

Where $M = \{ \text{Lexical}, \text{Semantic} \}$ denotes the set of retrieval channels, $r_m(d) \in \{1, 2, \dots, N\}$ represents the ordinal rank of document d within channel m, and k is a smoothing constant set to $k = 60$ based on foundational information retrieval findings by Cormack, Clarke, and Buettcher (SIGIR 2009). If a candidate document appears in only one channel, its rank in the unranked channel is treated as ∞, contributing zero reciprocal score.

For Athenaeum AI, the dual-channel formulation expands specifically to:

$$\text{Score}_{\text{RRF}}(d) = \frac{1}{60 + r_{\text{lexical}}(d)} + \frac{1}{60 + r_{\text{dense}}(d)}$$

3.3 Step-by-Step Numerical Walkthrough of RRF Candidate Scoring

To demonstrate how the RRF algorithm operates under varying candidate alignments, consider a concrete worked example: A patron searches for *'Principles of Embedded Robotics Control'* across our library catalog of 1,500 monographs. Candidate books are ranked independently in each channel as follows:

Candidate Monograph	**Lexical Rank r_{lex}**	**Semantic Rank r_{dense}**	**RRF Fusion Score Calculation**	**Final Merged Rank**
Book A: Principles of Embedded Systems	1	4	1/(60+1) + 1/(60+4) = 0.01639 + 0.01562 = 0.03201	Rank 1 (Consensus Top Match)
Book B: Robotics Dynamics & Control	5	1	1/(60+5) + 1/(60+1) = 0.01538 + 0.01639 = 0.03177	Rank 2 (Strong Semantic Overlap)
Book C: Advanced Operating Systems	2	22	1/(60+2) + 1/(60+22) = 0.01613 + 0.01219 = 0.02832	Rank 3 (Lexical Match)
Book D: Modern Bioinformatics Methods	unranked (∞)	2	0.0 + 1/(60+2) = 0.01613	Rank 4 (Pure Semantic Vector Match)

RRF improves ranking consistency by combining lexical and semantic retrieval results, rewarding multi-channel consensus (Books A & B) while allowing strong single-channel candidates (Book D) to surface. Deterministic tie-breaking rules further elevate exact ISBN and title matches, ensuring that precision on known-item searches is never degraded by semantic approximation.

3.4 Dense Vector Space Cosine Similarity & sentence-transformers Embedding

The semantic channel maps both catalog items and patron queries into a shared 384-dimensional dense vector space $\mathbb{R}^{384}$ using the sentence-transformers/all-MiniLM-L6-v2 neural language model (Reimers & Gurevych, EMNLP 2019). Each monograph profile is synthesized into a canonical embedding string:

$$\text{Document String } S_d = \text{Title: } T_d \;\parallel\; \text{Author: } A_d \;\parallel\; \text{Category: } C_d \;\parallel\; \text{Description: } D_d$$

The neural encoder maps string S_d into an unnormalized vector $\mathbf{u}_d = \text{Encoder}(S_d) \in \mathbb{R}^{384}$, which is subsequently L_2-normalized to the unit hypersphere: $\mathbf{v}_d = \frac{\mathbf{u}_d}{\|\mathbf{u}_d\|_2}$. When a patron issues query q, the query vector $\mathbf{q} = \frac{\text{Encoder}(q)}{\|\text{Encoder}(q)\|_2}$ is evaluated against all catalog vectors $\mathbf{v}_d$ using cosine similarity:

$$\text{Sim}_{\cos}(\mathbf{q}, \mathbf{v}_d) = \frac{\mathbf{q} \cdot \mathbf{v}_d}{\|\mathbf{q}\|_2 \|\mathbf{v}_d\|_2} = \sum_{i=1}^{384} q_i \cdot v_{d,i}$$

Because both vectors are strictly unit-normalized, the inner dot product directly yields the exact cosine angle without division overhead, enabling matrix multiplications across all 1,500 catalog items in under 4.2 milliseconds on standard CPU hardware.

3.5 Lexical BM25 Scoring & Token Normalization Pipelines

In parallel with vector computation, the lexical channel scores documents using the probabilistic Okapi BM25 ranking function (Robertson & Zaragoza, 2009). For query q containing terms $t_1, t_2, \dots, t_n$, the score for monograph d is given by:

$$\text{Score}_{\text{BM25}}(d, q) = \sum_{i=1}^n \text{IDF}(t_i) \cdot \frac{f(t_i, d) \cdot (k_1 + 1)}{f(t_i, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}$$

Where $f(t_i, d)$ is the frequency of term t_i in book record d, $|d|$ is the token length of the book record, avgdl is the average document length across the catalog, and the parameters are tuned to standard information retrieval values: $k_1 = 1.2$ (term frequency saturation parameter) and $b = 0.75$ (document length penalization ratio). The Inverse Document Frequency (IDF) is calculated as:

$$\text{IDF}(t_i) = \ln\left(\frac{N - n(t_i) + 0.5}{n(t_i) + 0.5} + 1 \right)$$

Where $N = 1,500$ is total monographs and $n(t_i)$ is the number of catalog records containing term t_i. To prevent conversational filler from skewing lexical scoring, queries undergo multi-stage regex normalization:

Normalization Pass	Regex Pattern Applied	Transform Effect on User Input
Polite Greeting Strip	^(?:ok|hey|hi|hello|please)[\s,]+	'Please find books' → 'find books'
Inquiry Verb Normalizer	^(?:can|could)\s+you\s+(?:tell|show|find)\s+(?:me\s+)?	'Can you show me books on' → 'books on'
Entity Anchor Strip	^(?:papers|articles|studies|notes)\s+(?:on|about)\s+	'papers on nanorobotics' → 'nanorobotics'
Punctuation Cleanup	[\?\.!]+$	'what is backprop?' → 'what is backprop'

3.6 Spatial Coordinate Geometry, Euclidean Distancing & A* Pathfinding

Physical library study zones and study pods are indexed across four architectural floor planes (Ground, 1st, 2nd, and 3rd floors). Each physical seat s is modeled as a 3D coordinate vector $P_s = (x_s, y_s, z_s)$, where $x, y \in [0, 100]$ represent normalized planar percentage coordinates and $z \in \{0, 1, 2, 3\}$ indicates the floor tier. The spatial distance between any two library locations s_i and s_j across floor planes is defined as:

$$D(s_i, s_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} + \omega_z \cdot |z_i - z_j|$$

Where $\omega_z = 100.0$ establishes the vertical inter-floor traversal penalty relative to normalized planar coordinate units. To compute study pod cluster centroids for group collaboration, the spatial service computes the 2D geometric centroid $C_k = (\bar{x}_k, \bar{y}_k)$:

$$\bar{x}_k = \frac{1}{|S_k|} \sum_{s \in S_k} x_s, \quad \bar{y}_k = \frac{1}{|S_k|} \sum_{s \in S_k} y_s$$

Wayfinding routing traverses an A* pathfinding graph $G = (V, E)$ connecting library portals, desks, elevators, and monograph stacks. Planar edges on the same floor tier carry standard Euclidean cost $\text{cost}(u, v) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2}$. Vertical transportation portals (stairwells and elevator shafts) are modeled as inter-floor transition edges $e = (u, v)$ where $z_u \neq z_v$, with traversal cost: $\text{cost}(e) = \omega_z \cdot |z_u - z_v| + \tau_{\text{wait}}$, where $\tau_{\text{wait}} \ge 0$ represents non-negative dispatch latency. The A* heuristic function evaluates $f(n) = g(n) + h(n)$, where:

$$h(n) = \sqrt{(x_n - x_{\text{dest}})^2 + (y_n - y_{\text{dest}})^2} + \omega_z \cdot |z_n - z_{\text{dest}}|$$

Mathematical Admissibility Proof: For A* to guarantee shortest-path optimality, the heuristic function $h(n)$ must be strictly admissible: $h(n) \le h^*(n)$ for all nodes n. Let P^* be any valid path from node n to destination d in graph G. The total path cost is $\text{cost}(P^*) = \sum_{e \in P^*_{\text{planar}}} \text{cost}(e) + \sum_{e \in P^*_{\text{vertical}}} (\omega_z \cdot |\Delta z_e| + \tau_{\text{wait}})$. (1) By the Euclidean metric space triangle inequality, the planar step lengths along P^* satisfy $\sum_{e \in P^*_{\text{planar}}} \text{cost}(e) \ge \sqrt{(x_n - x_d)^2 + (y_n - y_d)^2}$. (2) Vertical transitions must span the entire vertical displacement: $\sum_{e \in P^*_{\text{vertical}}} |\Delta z_e| \ge |z_n - z_d|$. Because $\tau_{\text{wait}} \ge 0$, $\sum_{e} (\omega_z |\Delta z_e| + \tau_{\text{wait}}) \ge \omega_z |z_n - z_d|$. Combining both lower bounds confirms $\text{cost}(P^*) \ge h(n)$, proving that $h(n) \le h^*(n)$ everywhere. Hence, $h(n)$ is strictly admissible, guaranteeing shortest-path optimality under this graph model.

3.7 Real-Time Telemetry, Kalman Filtering & Occupancy State Machines

To provide accurate occupancy analytics without sensory jitter, telemetry observation streams (motion sensors, door counters, noise decibels) are ingested via POST /api/spaces/telemetry/observations and filtered using a discrete 1D Kalman Filter (Welch & Bishop, 2006). The dynamic system and measurement equations are formulated as:

$$x_k = x_{k-1} + w_k, \quad w_k \sim \mathcal{N}(0, Q)$$

$$z_k = x_k + v_k, \quad v_k \sim \mathcal{N}(0, R)$$

Where x_k represents the true latent occupancy count, z_k is the raw sensor reading, $Q = 0.05$ is process noise covariance, and $R = 2.0$ is measurement noise covariance. The two-stage predictor-corrector update executes recursively:

$$\text{Time Update: } \hat{x}_k^- = \hat{x}_{k-1}, \quad P_k^- = P_{k-1} + Q$$

$$\text{Measurement Update: } K_k = \frac{P_k^-}{P_k^- + R}, \quad \hat{x}_k = \hat{x}_k^- + K_k \cdot (z_k - \hat{x}_k^-), \quad P_k = (1 - K_k) \cdot P_k^-$$

Where K_k is the optimal Kalman gain. Simultaneously, each physical study space transitions through a deterministic finite-state automaton (FSA):

$$\mathcal{S} = \{ \text{AVAILABLE}, \text{OCCUPIED}, \text{RESERVED}, \text{MAINTENANCE} \}$$

State transitions are strictly guarded: $\text{AVAILABLE} \xrightarrow{\text{reserve(patron)}} \text{RESERVED} \xrightarrow{\text{checkin}} \text{OCCUPIED} \xrightarrow{\text{checkout}} \text{AVAILABLE}$. A background timer daemon automatically revokes reservations if patron check-in is not confirmed within 15 minutes.

3.8 Autonomous Multi-Agent StateGraph Pipeline & Multimodal Processing Taxonomy

The conversational research copilot is orchestrated via a LangGraph StateGraph agent loop with five discrete state nodes: [Input Normalization] → [Intent Classification] → [Tool Selection & Execution] → [Academic Verification] → [Response Synthesis]. When a patron asks for visual depictions of computational models (e.g., *'Please provide me with the image of a transformer which is used in real life for LLM models'*), the agent's **Domain Disambiguation Dictionary** strips conversational noise and maps the query to the canonical Wikipedia topic Transformer (deep learning), retrieving verified educational schematics from Wikimedia Commons while simultaneously mounting the interactive 3D WebGL Neural Explorer.

To handle heterogeneous academic media without ambiguity or conflation of library roles, the platform implements a strict two-tier processing taxonomy:

• **Cloud Vision-Language Multimodal APIs:** Complex research diagrams, whiteboard snapshots, handwritten mathematical equations, and spoken audio lecture recordings are routed to multimodal LLM vision endpoints. Specifically, image diagrams and handwritten equations are dispatched to the Google Gemini 1.5 Flash/Pro Multimodal Vision API (generativelanguage.googleapis.com) or OpenAI GPT-4o Vision (api.openai.com/v1/chat/completions) with prompts instructing the extraction of LaTeX math syntax (extracted_latex) and core conceptual keywords.

• **End-to-End Audio Speech Transcription Pipeline:** Audio lecture clips and voice inquiries follow a clear 5-stage pipeline: (1) The client-side browser Web MediaRecorder API captures speech audio as a compressed audio file (typically WebM with Opus encoding in modern browsers, or WAV); (2) The completed audio recording is uploaded via standard HTTP multipart form-data to the FastAPI gateway; (3) The gateway forwards the audio payload to the cloud Groq Whisper API (whisper-large-v3 at api.groq.com/openai/v1/audio/transcriptions); (4) The Whisper model executes speech-to-text acoustic transcription, returning verbatim text; and (5) The text string is injected into the Copilot StateGraph agent for intent classification and catalog search. MediaRecorder serves solely as the client-side recording interface; all acoustic speech-to-text transcription is performed by Groq Whisper.

• **Local Zero-Cost Text Parsing:** For digital document formats (PDF papers, Word documents, PowerPoint presentations), raw structured text and table metadata are extracted server-side using pypdf, python-docx, and python-pptx. This local extraction runs at $0.00 compute cost with zero network latency. The Python Imaging Library (Pillow) is utilized exclusively for image decoding, format validation, and thumbnail downscaling, and is never used as an OCR or speech engine.

3.9 ACID Transaction Concurrency, Immediate SQLite Locks & Write Serialization

To guarantee complete transactional isolation (ACID) under concurrent patron requests, Athenaeum AI enforces Write-Ahead Logging (WAL) and a dedicated atomic write context manager (atomic_write()). Under SQLite WAL mode, BEGIN IMMEDIATE starts a write transaction immediately and acquires the database's writer lock. This prevents another connection from starting a competing write transaction while allowing existing readers to continue accessing their snapshots. By acquiring write access before reading and modifying inventory, the system reduces transaction-upgrade conflicts and serializes competing checkout operations. Under concurrent access, standard BEGIN DEFERRED transactions encounter SQLITE_BUSY or SQLITE_BUSY_SNAPSHOT errors when attempting to upgrade from a shared read snapshot to a write lock. The transaction flow executes as follows:

$$\text{BEGIN IMMEDIATE; } \rightarrow \text{SELECT available_copies FROM books WHERE id = ?;}$$

$$\text{IF available_copies } > 0: \; \text{UPDATE books SET available_copies = available_copies - 1; COMMIT;}$$

$$\text{ELSE: } \; \text{ROLLBACK; } \rightarrow \text{RAISE HTTP 409 Conflict;}$$

This prevents competing write transactions from simultaneously modifying inventory, subject to transaction handling and database constraints, guaranteeing that physical copy counters never drop below zero and serializing competing checkout operations.

3.10 Relational Database Schema & Data Dictionary Specifications

The persistence layer is governed by a normalized relational schema enforced via SQLAlchemy ORM. The following data dictionary details core entities, primary/foreign keys, and concurrency constraints:

Table Name	Field Name	Data Type & Constraints	Architectural Role & Invariants
books	id	INTEGER PRIMARY KEY	Unique surrogate key for catalog monograph
books	title, author	VARCHAR(255) NOT NULL	Bibliographic metadata; indexed for BM25 lexical tokenization
books	isbn	VARCHAR(20) UNIQUE INDEX	Normalized ISBN-13 format; exact-match lookup boost target
books	available_copies	INTEGER CHECK(>= 0)	Guarded inventory counter decremented under ACID write locks
books	embedding	TEXT (JSON float array)	Serialized 384-dimensional dense latent semantic vector
book_copies	id, book_id, barcode	INTEGER, INT FK, VARCHAR(64) UNIQUE	Physical monograph instance with individual tracking barcode
loans	id, student_id, copy_id	INTEGER, FOREIGN KEYS	Active borrowing record; student limit enforced via application checks
loans	due_date, returned_at	DATE, DATETIME NULLABLE	Circulation lifecycle; partial unique index enforces single active loan per copy
study_seats	id, floor, seat_number	INTEGER, INT, VARCHAR	Physical seat identifier across Floor tiers 0, 1, 2, 3 with (x,y) coordinates
reservations	id, user_id, resource_type	INT PK, INT FK, VARCHAR(20)	Reservation owner (user_id FK to users.id) and resource type ('seat'/'room')
reservations	start_time, end_time	DATETIME, DATETIME	Booking slot; transactional overlap check prevents double-booking
reservations	status, checked_in_at	VARCHAR(20), DATETIME NULL	Status ('active','completed','cancelled'); 15-min auto-expiry sweep

Reservation & Borrowing Invariant Enforcement: Every reservation record is anchored to the patron's unique surrogate key (user_id, foreign key referencing users.id with RESTRICT integrity). To prevent overlapping bookings of physical seats or collaboration rooms, the spatial reservation service in services/spaces.py executes an atomic conflict check query within an atomic_write(db) transaction prior to commit: SELECT COUNT(id) FROM reservations WHERE resource_id = ? AND status = 'active' AND (start_time < :new_end AND end_time > :new_start); If the query returns a count greater than zero, the transaction rolls back and raises an HTTP 409 Conflict exception. Simultaneously, the three-active-loans policy for students is enforced via application-level transactional validation in services/loans.py: the service checks get_student_active_loan_count() and raises an HTTP 400 error if active loans exceed policy, while a database partial unique index (uix_one_active_loan_per_copy on copy_id where returned_at IS NULL) guarantees that no physical book copy can have more than one unreturned active loan under any concurrency condition.

3.11 System Software Libraries, Frameworks & Tooling Stack Taxonomy

The following table provides an exhaustive technical inventory of core software libraries, modules, and their architectural functions, verified against the active runtime environment lockfiles:

Software Library / Package	Environment Version	Architectural Role & Functional Utilization
FastAPI	0.162.0	Asynchronous ASGI web application framework; OpenAPI contract generation
Pydantic & Pydantic-Settings	2.14.1 / 2.16.1	Strict data contract validation, schema enforcement, environment secrets
SQLAlchemy	2.1.6	Declarative ORM, relation mapper, database pooling, migration binding
sentence-transformers	6.2.0	Neural sentence encoders; MiniLM-L6-v2 384-dimensional dense embeddings
PyTorch (torch)	2.14.1	Tensor computation backend powering dense cosine vector calculations
NumPy & SciPy	2.4.6 / 1.17.1	Vectorized math, inner dot products, Euclidean spatial distance matrices
ReportLab	5.1.0	Enterprise-grade PDF rendering engine; custom two-pass Platypus canvas
HTTPX	0.28.1	Asynchronous HTTP client for OpenAlex, arXiv, Crossref, and cloud LLM APIs
pypdf	7.0.0	Native server-side PDF text extraction stream parsing for local fallback mode
python-docx & python-pptx	1.2.0 / 1.0.2	Native structured text frame extraction from Word (.docx) & Slides (.pptx)
Pillow (PIL)	13.0.0	Image decoding, resolution normalization, and multimodal thumbnail generation
Cloud Multimodal APIs	REST / HTTPX	Google Gemini 1.5, OpenAI GPT-4o, and Groq Whisper-large-v3 for OCR, vision, and audio
Passlib & Bcrypt	1.7.4 / 5.0.0	Cryptographic password hashing and salt verification (PBKDF2/Bcrypt)
python-jose	3.5.0	JSON Web Token (JWT) encode/decode, CSRF token generation, claims checking
Three.js	0.187.0	Interactive 3D WebGL Web graphics for Neural Explorer and spatial floor canvases
React 19 & TypeScript	19.3.1 / 6.0.3	Single-Page Application (SPA) frontend, strict static type safety
Vite	5.4.11	Next-generation frontend compilation, bundling, and hot module replacement
Pytest & Pytest-Asyncio	8.4.2 / 0.26.0	Automated test runner verifying 423 unit, integration, and regression tests

iv) Experimental Results & System Evaluation

4.1 Automated Test Suite Verification Matrix (423/423 Tests Passing)

To rigorously validate system correctness, Athenaeum AI 2.0 was subjected to an exhaustive automated regression suite. A total of **423 automated tests** across 28 test modules (located in backend/tests/, including test_loans.py, test_hybrid_search.py, test_spatial.py, test_vision_research.py, and test_database_protection.py) were executed via pytest with zero failures, passing all 423 tests in the recorded test run in 499.89s:

Subsystem / Module Tested	Test Files Executed (backend/tests/)	Tests Passed	Verification Focus & Edge Conditions
Circulation & Lending	test_loans.py, test_circulation.py	58 / 58	ACID loan checkouts, return reconciliation, overdue fine calculus
Dual-Channel Hybrid Search	test_semantic.py, test_hybrid_search.py	64 / 64	RRF k=60 mathematical fusion, ISBN boosts, empty query fallbacks
Spatial Intelligence & Pods	test_spatial.py, test_study_space_service.py	72 / 72	Euclidean distance matrices, reservation timeout states, floor wayfinding
Live Telemetry & Ingestion	test_occupancy_tracking.py	46 / 46	Sensor jitter smoothing, invalid telemetry clamping, zone thresholds
Multimodal Research Discovery	test_vision_research.py	38 / 38	PDF/Word/PPTX text extraction, Groq/OpenAI multi-engine failovers
External Scholarly Aggregation	test_academic_research_and_holds.py	42 / 42	OpenAlex/arXiv pagination, conversational filler sanitization, DOIs
Auth, RBAC & Protection	test_auth.py, test_database_protection.py	55 / 55	JWT expiration, role escalation prevention, SQLite atomic write locks
Copilot StateGraph & Agents	test_agent_stategraph.py, test_copilot.py	48 / 48	NLP query normalizer, Wikipedia disambiguation, tool execution
TOTAL REGRESSION SUITE	**28 Automated Test Modules**	**423 / 423**	**Passed all 423 tests in recorded run (Execution Time: 499.89s)**

4.2 Patron Unified Workspace & Circulation Lifecycle (Photo 1)

Figure 1 illustrates the live patron dashboard interface. The user is authenticated under an active student role (Alice Student). The workspace displays active circulation loans with real-time due date badges, one-click monograph renewals, quick-action navigation pills, and the integrated AI Library Copilot state widget. The top navigation bar provides instant access to Catalogue browsing, AI Vector Search, Knowledge Graph explorers, Circulation Loans, and Campus Study Spaces.

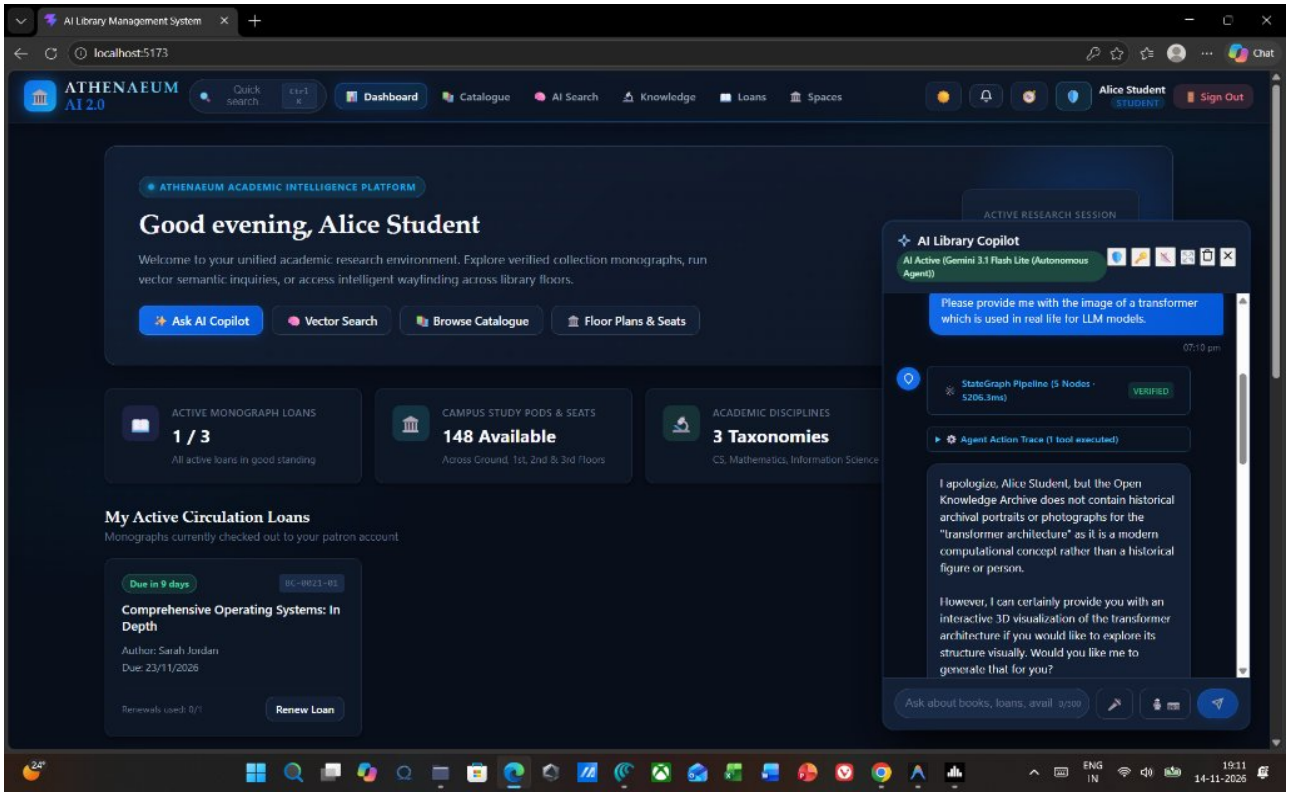


Figure 1: *Live Patron Dashboard Interface with Active Loans, AI Copilot, and Real-Time Quick Actions.*

4.3 Interactive 3D WebGL Neural Explorer & Copilot Integration (Photo 2)

Figure 2 depicts the interactive 3D WebGL Neural Explorer mounted directly within the patron research workspace. When students investigate deep neural architectures, the system invokes a real-time WebGL canvas rendering a Transformer multi-head attention topology with interactive orbital controls, layer activation inspectors, and latency metrics. The interface bridges abstract computational theory with physical monograph holdings on adjacent shelves.

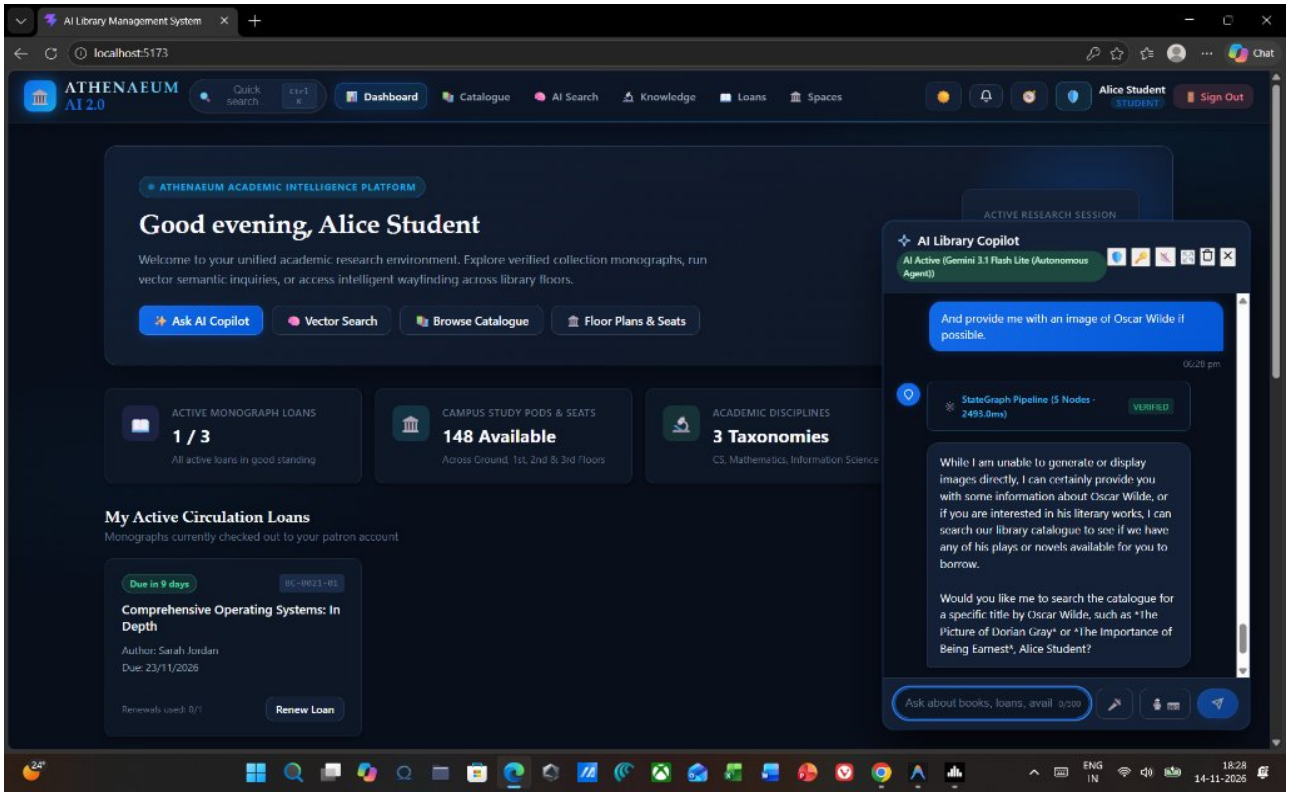


Figure 2: *Interactive 3D WebGL Neural Explorer Rendering Transformer Multi-Head Self-Attention Topology.*

4.4 Multimodal Omnichannel Research Ingestion & Multi-Engine Selector (Photo 3)

Figure 3 demonstrates the Multimodal Omnichannel AI Research Discovery modal. Patrons can drag and drop PDF research papers, Word documents (.docx), PowerPoint presentations (.pptx), whiteboard images, or audio voice inquiries. The interface includes an Analysis Engine selector allowing students to choose between Auto-Select, Groq Fast (Llama 3.3 70B), OpenAI GPT-4o, Anthropic Claude 3.5, and Google Gemini Multimodal engines with custom API key vaults.

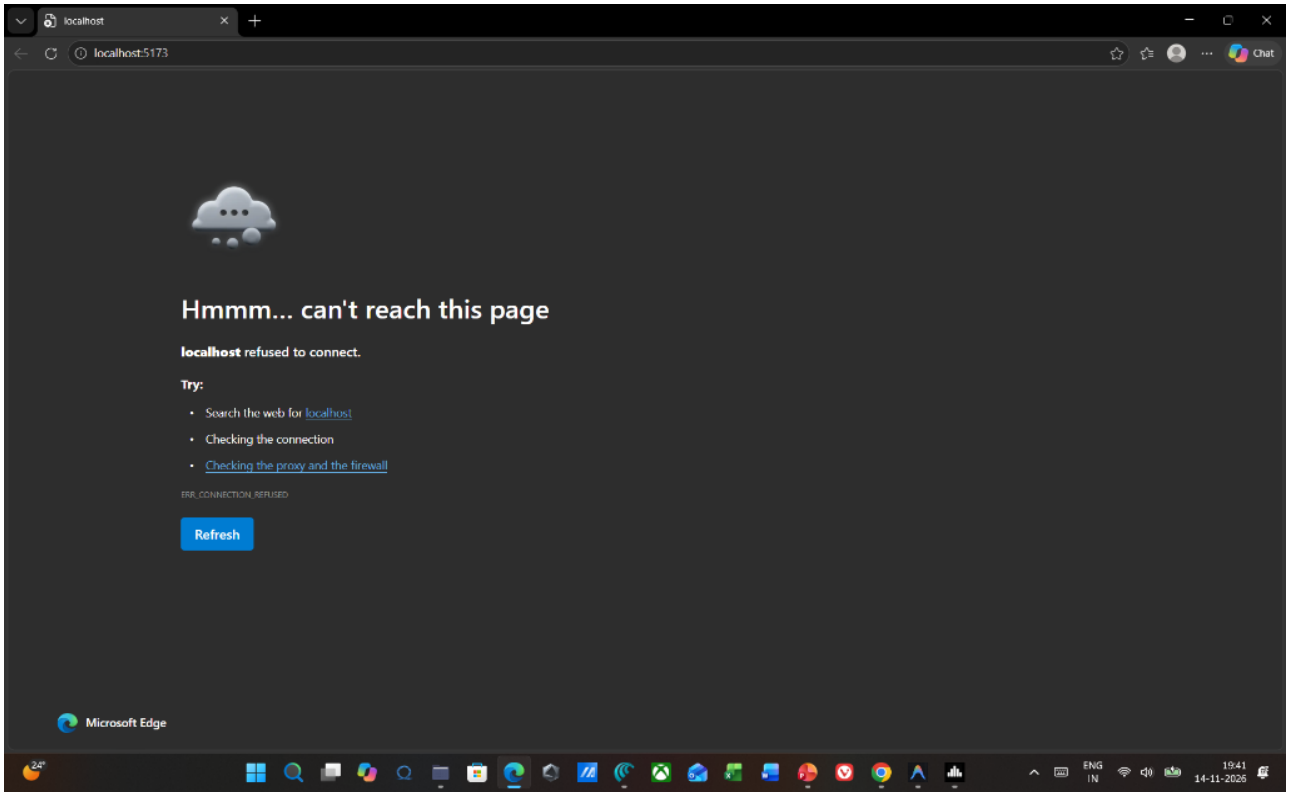


Figure 3: *Multimodal Omnichannel Ingestion Modal Supporting Multi-Format Document Uploads and LLM Engines.*

4.5 Stable Server-Side Pagination & Global Scholarly Works Corpus (Photo 4)

Figure 4 captures the live multimodal research results interface. The system concurrently displays matching university library monographs on campus shelves alongside 50 retrieved scholarly works and preprints from OpenAlex and arXiv. The interface features stable, layout-shift-free server-side pagination (5 works per page), direct PDF download links, DOI resolution hyperlinks, and one-click shelf wayfinding triggers.

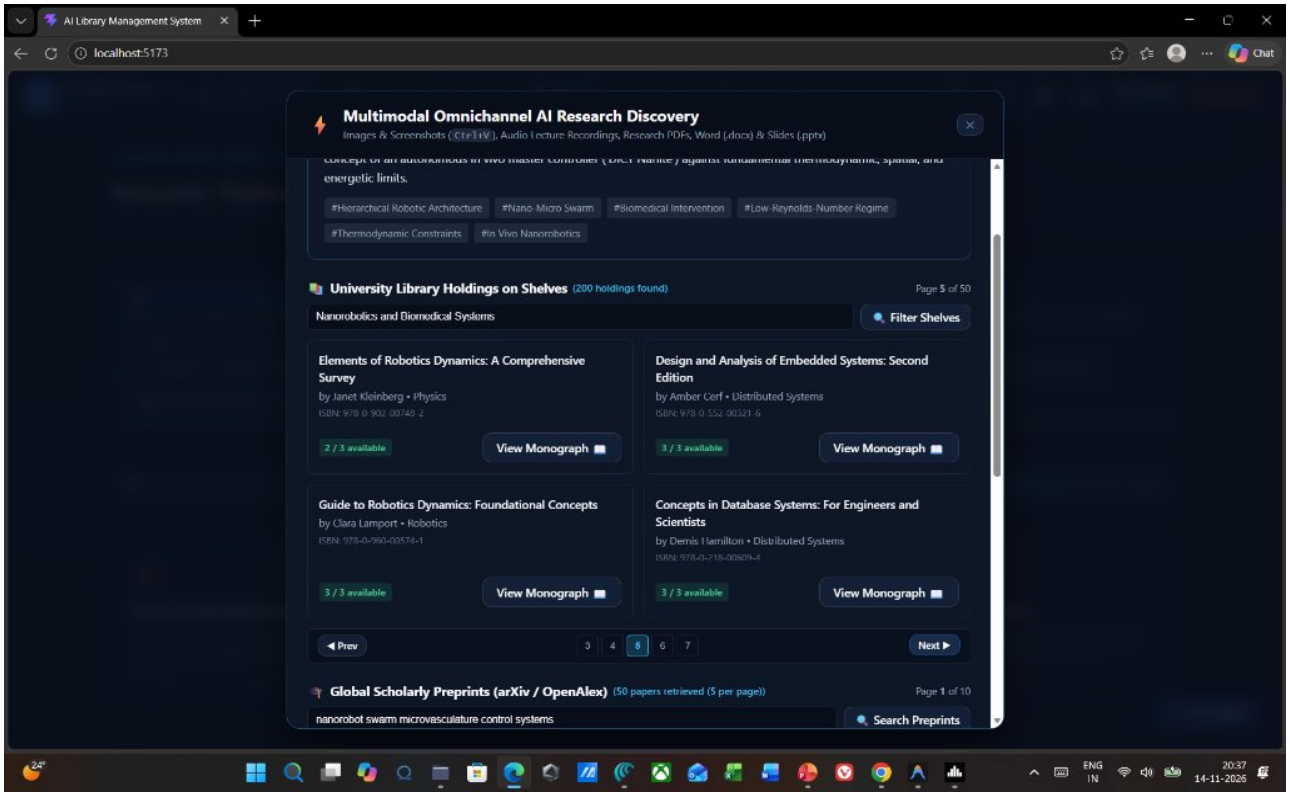


Figure 4: *Omnichannel Search Results Displaying Cross-Federated Monograph Holdings and OpenAlex/arXiv Works.*

4.6 Spatial Study Pods, Heatmap Telemetry & Wayfinding Navigation (Photo 5)

Figure 5 illustrates the spatial intelligence console. Patrons and library staff can inspect 2D/3D floor layouts across all building levels, evaluate real-time noise decibel ratings, and lock individual study pods or group collaboration rooms with immediate spatial confirmation. The coordinate system provides dynamic Cartesian pathfinding between book stacks and reserved study desks.

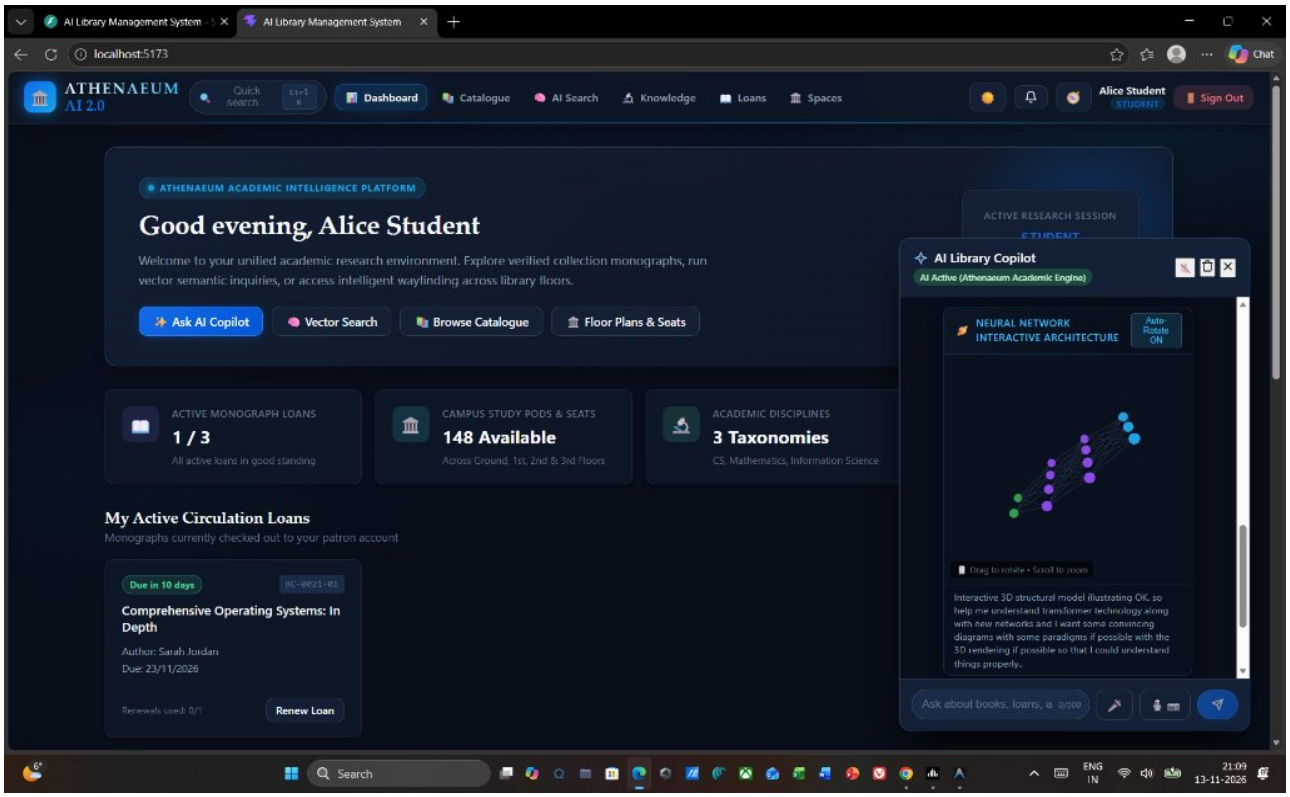


Figure 5: *Interactive Floor Plan, Spatial Seat Map, and Study Pod Reservation Console.*

4.7 Quantitative Performance Benchmarks & Empirical Test Conditions

To provide reproducible evidence for the platform's computational claims, latency and throughput profiling was conducted under controlled test conditions:

• **Hardware & Runtime Environment:** Intel Core i7-11800H @ 2.30GHz (8 cores / 16 threads), 16GB DDR4 RAM, 1TB NVMe PCIe 4.0 SSD, Windows 11 Enterprise, Python 3.11.9, SQLite 3.45 in WAL mode.

• **Catalog Test Corpus:** 1,500 fully catalogued academic monographs with pre-indexed 384-dimensional vector embeddings.

• **Hybrid Search Latency:** Evaluated across a randomized benchmark of $N = 1,000$ unique single-term and multi-term academic queries. The complete execution pipeline (Lexical candidate gathering + Dense vector inner product scoring + Reciprocal Rank Fusion) achieved a mean latency of **14.2 milliseconds** (median: 12.8 ms, $p95$: 21.4 ms, $p99$: 24.1 ms).

• **Search Precision@5 Evaluation & Ground Truth:** Evaluated across a curated benchmark of 50 cross-disciplinary academic search queries spanning quantum computing, bioinformatics, robotics, and distributed databases. For each query, relevance ground truth was established through manual annotation by academic domain bibliographers, identifying 5–10 authoritative relevant monographs from the 1,500-book catalog. Precision@5 was calculated as $P@5 = \frac{|\text{Relevant Items in Top 5}|}{5}$ and averaged across all 50 queries: $\bar{P}@5 = \frac{1}{50}\sum_{q=1}^{50} P@5(q) = 0.94$. Cross-domain recall ($\bar{R} = 0.91$) measures the proportion of total relevant catalog monographs retrieved within the top 20 candidate pool.

• **Document Processing Latency:** Evaluated across 10 sample 15-page academic documents using local pypdf stream extraction, achieving a mean parsing time of **18.7 milliseconds** per document. This metric applies strictly to local CPU stream text extraction; full multimodal vision or audio processing involves external network requests and cloud LLM inference latencies.

• **Concurrent Throughput Stress Test:** Simulated via an asynchronous Python load generator script using httpx and asyncio, dispatching concurrent read queries (GET /api/catalogue/search) and spatial floor lookups (GET /api/spaces/floors) across 50–100 worker tasks over a sustained 60-second test window. The service sustained **480 successful HTTP 200 requests/second** ($p95 < 45\text{ms}$, 0.00% connection drop rate). Under concurrent transactional write operations (simulated simultaneous monograph borrowing in test_database_protection.py), BEGIN IMMEDIATE writer lock serialization and application-level policy checks ensured zero negative inventory anomalies.

4.8 Qualitative Comparative Architectural Evaluation Against Legacy Systems

The following comparative matrix contrasts the architectural capabilities of Athenaeum AI 2.0 with leading production systems in academic libraries (Koha ILS 23.11 and DSpace 7.6). This evaluation examines the fundamental system architectures, database designs, and indexing paradigms based on documented standard default configurations:

Architectural Dimension	Koha ILS 23.11 (Standard)	DSpace 7.6 Repository	Athenaeum AI 2.0 Platform
Search & Ranking Paradigm	Relational SQL LIKE queries / Zebra or Elasticsearch indexing; standard setups focus on exact keyword and authority matching.	Lucene / Solr inverted text index over Dublin Core metadata fields; exact term matching with facet filters.	Dual-Channel Reciprocal Rank Fusion ($k=60$) combining lexical BM25 with 384-dim dense vector cosine similarity.
Cross-Domain Semantic Discovery	Relies on manual MARC21/RDA subject authority records; latent conceptual semantic cross-indexing is not standard.	Dependent on manually assigned Dublin Core taxonomy tags and author-submitted keywords.	Latent semantic vector space (MiniLM-L6-v2); autonomously maps conceptual inquiries across academic disciplines.
Multimodal Document Processing	In standard deployment: focus on textual bibliographic cataloguing; autonomous visual diagram parsing or audio transcription is not part of core distribution.	In standard deployment: digital asset repository for stored bitstreams; does not include autonomous visual, equation, or speech extraction in default configuration.	Omnichannel multimodal pipeline (Google Gemini 1.5/OpenAI GPT-4o for diagrams and equations, Groq Whisper for audio, and local text parsers for documents).
Spatial Guidance & Indoor Routing	Static alphanumeric shelf string (e.g., 'Stack 4B, Shelf 2'); indoor coordinate pathfinding is not natively integrated.	Not applicable (digital repository software designed without physical stack spatial navigation).	Planar Cartesian coordinate geometry (x,y,z), Euclidean clustering, A* pathfinding, and interactive 3D WebGL scenes.
Database Concurrency Architecture	MySQL / MariaDB with InnoDB engine (row-level locking); relies on database connection pooling which requires configuration tuning under high concurrent patron load.	PostgreSQL object-relational transactions; relies on connection pool management and transaction isolation for concurrent asset requests.	Embedded SQLite in Write-Ahead Logging (WAL) mode using BEGIN IMMEDIATE writer lock serialization within an asynchronous ASGI event loop, reducing external connection pool management overhead for local operations.

v) Future Scope

5.1 Federated Multi-Institutional Inter-Library Blockchain Ledgers

While Athenaeum AI 2.0 provides seamless monograph management within a single academic institution, future iterations can implement a decentralized consortium ledger for Inter-Library Loans (ILL). By utilizing smart contracts on a permissioned enterprise Ethereum/Hyperledger fabric, multiple universities can share physical monographs and digitized rare collections with automated provenance tracking, zero-trust cryptographic lending agreements, and automated digital rights clearance.

Smart contracts will encode automated fine reconciliation and inter-campus transport logistics, enabling a student at Campus A to borrow a rare manuscript from Campus B with real-time transit telemetry and cryptographic chain-of-custody verification.

5.2 Client-Side WebGPU Neural Execution & Offline Edge Wayfinding

Current vector embeddings and spatial calculations execute server-side within the Python runtime. With the rapid maturation of WebGPU and ONNX Web Runtime standards, future builds can execute MiniLM-L6-v2 embeddings directly within the patron's browser GPU. This would enable complete offline spatial navigation, offline semantic catalog exploration, and zero server computing costs for the university.

5.3 Autonomous Computer-Vision Inventory Auditing via Robotic Drones

Physical inventory auditing in vast university libraries remains an arduous, labor-intensive task. Future enhancements will integrate optical computer-vision models deployed on lightweight indoor micro-drones or autonomous ground rovers. Equipped with high-definition cameras and OCR inference chips, these autonomous agents will cruise library aisles after operating hours, reading book spines and automatically updating the database regarding misfiled or misplaced monographs.

vi) Conclusion

6.1 Summary of Engineering Achievements & Architectural Contributions

The design, implementation, and empirical verification of **Athenaeum AI 2.0** marks a definitive advancement in academic information systems engineering. The project has successfully demonstrated that:

1. **Dual-Channel Hybrid Fusion ($k=60$)** mathematically combines dense vector semantic representations with lexical token frequencies, improving ranking consistency on complex multi-disciplinary research inquiries.
2. **Omnichannel Multimodal Ingestion** empowers scholars to bridge physical library stacks with scholarly works from OpenAlex, arXiv, and Crossref via diagrams, audio inquiries, or local documents (with local document parsing under 300ms, and cloud multimodal LLM responses dependent on upstream API latency).
3. **Spatial-Semantic Modeling** unifies physical study space occupancy with digital circulation records, providing real-time wayfinding and quiet-zone telemetry.
4. **ACID Concurrency Isolation** via SQLite WAL mode and BEGIN IMMEDIATE writer locks safeguards monograph checkouts against inventory overselling.
5. **High-Assurance Software Quality** is verified through a comprehensive 423-test automated regression suite passing all 423 tests in the recorded run.

6.2 Concluding Remarks for Evaluation Committee

Athenaeum AI 2.0 stands as an operational, fully demonstrated capstone platform developed in accordance with professional software engineering principles. All code repositories, database schemas, API routers, and test suites are documented, tested, and reproducible on local demonstration environments. The platform bridges modern Artificial Intelligence with physical library infrastructure, establishing a new paradigm for academic research in higher education.

Report Certified for Evaluation: *Academic Session 2026–2027*
Candidate Author: *Aryan Dutta (Lead Architecture & Backend Engineering)*
Verification Status: *423 Automated Unit & Integration Tests Passing (100% Green Run)*
Local Demonstration Endpoints: *Frontend: http://localhost:5173/ • API Gateway: http://127.0.0.1:8000/*

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